آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Pacing Guide for Teachers

PHYSICS

Grade XI

Theory

FOR ACADEMIC YEAR 2025 AND ONWARDS

Number of weeks: 28

Number of periods per week: 5

Key Textbook: Model Textbook of Physics 11, National Book Foundation, Islamabad (Based on National Curriculum of Pakistan 2022-2023)

Teacher Developer(s) and their Institution(s):

Muhammad Ali Rana - Ghulaman-e-Abbas School, Karachi

Israr-ul-Haq - Ghulaman-e-Abbas School, Karachi

Saib Ahmed Khan - Govt. Degree Science and Commerce College, Orangi Township, Karachi

Qasim Abbas - Aga Khan Higher Secondary School, Kharadar, Karachi

Muhammad Affan Khan - Aga Khan Higher Secondary School, Karimabad, Karachi

Syeda Shazia Ahmed - Habib Girl's Higher Secondary School, Karachi

Muhammad Salman - Habib Public High School, Karachi

Topic	Total Periods
1. Physical Quantities and Measurement	08

Sub-Topic	Range of SLOs	Periods (40 mins)
1.1 Physical Quantities	1.1.1 – 1.1.5	02
1.2 Dimensions	1.2.1 – 1.2.4	02
1.3 Precision and Accuracy	1.3.1, 1.3.2	02
1.4 Significant Figures and Uncertainty	1.4.1, 1.4.2	02

Web Resource

https://circuitglobe.com/accuracy-and-precision.html

Suggested Activities and/ or Formative Assessment

Activity 1:

Measured values activity that includes uncertainty.

https://serc.carleton.edu/sp/library/uncertainty/examples/48732.html

Activity 2

Prepare a set of CRQ level questions in the form of chits, matching the number of questions to the total number of students in the class. Randomly distribute these chits among the students and instruct them to answer the questions in their own copies. After collecting the answers, assess and grade each student's response. To ensure the reusability of the questions for the following year, strictly prohibit students from writing anything on the chits during the exercise.

FOR ACADEMIC YEAR 2025 AND ONWARDS

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resources%20Guide%202025%20HSSC%20Physics.pdf

Further Resource

For additional resources related to teaching, learning and formative assessments, please refer to **Learn Smart Classroom by Knowledge Platform:**

https://akueb.knowledgeplatform.com/login

FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
2. Scalars and Vectors	11

Sub-Topic	Range of SLOs	Periods (40 mins)
2.1 Introduction to Vectors	2.1.1	02
2.2 Addition of Vectors by Rectangular Component Method	2.2.1 – 2.2.3	03
2.3 Scalar Product of Two Vectors	2.3.1 – 2.3.3	03
2.4 Vector Product of Two Vectors	2.4.1 – 2.4.3	03

Web Resource

https://openstax.org/books/university-physics-volume-1/pages/2-4-products-of-vectors

Suggested Activities and/ or Formative Assessment

Activity 1

Instruct the students to come up with one or two CRQ level questions either through their own thinking or by conducting research. Then, randomly distribute these questions among the students in the class. They can choose to solve the questions either individually or in groups during the class session. This activity encourages critical thinking and collaboration among the students while engaging them in answering challenging questions.

Activity 2

Few sample questions for assessments after completion of topics.

https://www.slideshare.net/RamC3/physics-notes-solved-numerical-of-physics-first-year

FOR ACADEMIC YEAR 2025 AND ONWARDS

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resources%20Guide%202025%20HSSC%20Physics.pdf

Further Resource

For additional resources related to teaching, learning and formative assessments, please refer to **Learn Smart Classroom by Knowledge Platform:**

https://akueb.knowledgeplatform.com/login

FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
3. Motion in One Dimension	11

Sub-Topic	Range of SLOs	Periods (40 mins)
3.1 Equations of Motion	3.1.1, 3.1.2	02
3.2 Principle of Conservation of Momentum	3.2.1 – 3.2.5	03
3.3 Work	3.3.1, 3.3.2	02
3.4 Conservative and Non-Conservative Fields	3.4.1	02
3.5 Work-Energy Theorem	3.5.1, 3.5.2	02

FOR ACADEMIC YEAR 2025 AND ONWARDS

Web Resources

https://openstax.org/books/university-physics-volume-1/pages/3-4-motion-with-constant-acceleration

https://www.cyberphysics.co.uk/topics/forces/momentum.htm

https://openstax.org/books/university-physics-volume-1/pages/4-3-projectile-motion

Suggested Activities and/ or Formative Assessment

Activity 1

Problem solving based Simulation Activity.
https://phet.colorado.edu/en/simulations/projectile-motion

Activity 2

Few sample questions for assessments after completion of topics.

https://www.slideshare.net/RamC3/physics-notes-solved-numerical-of-physics-first-year

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resources%20Guide%202025%20HSSC%20Physics.pdf

Further Resource

For additional resources related to teaching, learning and formative assessments, please refer to **Learn Smart Classroom by Knowledge Platform:**

https://akueb.knowledgeplatform.com/login

FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
4. Motion in Two Dimensions	12

Sub-Topic	Range of SLOs	Periods (40 mins)
4.1 Projectile	4.1.1 – 4.1.4	03
4.2 Angular Motion	4.2.1 – 4.2.3	02
4.3 Centripetal Force and Centripetal Acceleration	4.3.1 – 4.3.5	02
4.4 Moment of Inertia	4.4.1 – 4.4.4	02
4.5 Angular Momentum	4.5.1 – 4.5.4	03

FOR ACADEMIC YEAR 2025 AND ONWARDS

Web Resources

https://openstax.org/books/university-physics-volume-1/pages/8-3-conservation-of-energy

https://www.cyberphysics.co.uk/topics/energy/sources.htm

Suggested Activities and/ or Formative Assessment

Activity 1

Ask students to use this simulator and give tasks to accomplish about energy, types and conservation.

https://javalab.org/en/mechanical_energy_en/

Activity 2

Ask students to inculcate their ICT skills to design some quizzes e.g., Google form, Wiser, Quizzes, ProProfs etc. for some assigned topics.

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resources%20Guide%202025%20HSSC%20Physics.pdf

Further Resource

For additional resources related to teaching, learning and formative assessments, please refer to **Learn Smart Classroom by Knowledge Platform:**

https://akueb.knowledgeplatform.com/login

FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
5. Deformation of Solids	10

Sub-Topic	Range of SLOs	Periods (40 mins)
5.1 Types of Solids	5.1.1	02
5.2 Deformation in Solids	5.2.1	02
5.3 Tensile and Plastic Deformation	5.3.1	02
5.4 Stress, Strain and Young's Modulus	5.4.1 – 5.4.4	02
5.5 Elastic Potential Energy of Materials	5.5.1	02

FOR ACADEMIC YEAR 2025 AND ONWARDS

Web Resources

https://openstax.org/books/university-physics-volume-1/pages/4-4-uniform-circular-motion

https://openstax.org/books/university-physics-volume-1/pages/11-3-conservation-of-angular-momentum

https://openstax.org/books/university-physics-volume-1/pages/6-3-centripetal-force

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resources%20Guide%202025%20HSSC%20Physics.pdf

Further Resource

For additional resources related to teaching, learning and formative assessments, please refer to **Learn Smart Classroom by Knowledge Platform:**

https://akueb.knowledgeplatform.com/login

FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
6. Fluids Dynamics	14

Sub-Topic	Range of SLOs	Periods (40 mins)
6.1 Floating Objects	6.1.1 – 6.1.5	02
6.2 Streamline and Turbulent Flow	6.2.1, 6.2.2	02
6.3 Equation of Continuity	6.3.1 – 6.3.3	03
6.4 Bernoulli's Equation	6.4.1 – 6.4.3	03
6.5 Viscous Fluids and Fluid Friction	6.5.1 – 6.5.5	04

FOR ACADEMIC YEAR 2025 AND ONWARDS

Web Resources

https://www.princeton.edu/~asmits/Bicycle_web/continuity.html

https://www.youtube.com/watch?v=wykn-JTnacE

https://www.youtube.com/watch?v=bC8v6hlXnSk&t=13s

https://www.youtube.com/watch?v=p0lZsfzDS4s

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resources%20Guide%202025%20HSSC%20Physics.pdf

Further Resource

For additional resources related to teaching, learning and formative assessments, please refer to **Learn Smart Classroom by Knowledge Platform:**

https://akueb.knowledgeplatform.com/login

FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
7.Oscillations	18

Sub-Topic	Range of SLOs	Periods (40 mins)
7.1 Simple Harmonic Motion (S.H.M.)	7.1.1, 7.1.2	02
7.2 Uniform Circular Motion and S.H.M.	7.2.1 – 7.2.5	02
7.3 Mass-Spring System	7.3.1 – 7.3.3	03
7.4 Simple Pendulum	7.4.1 – 7.4.3	03
7.5 Energy Conservation in S.H.M.	7.5.1 – 7.5.3	03
7.6 Free and Forced Oscillation	7.6.1	01
7.7 Resonance	7.7.1, 7.7.2	02
7.8 Damped Oscillations	7.8.1 – 7.8.3	02

FOR ACADEMIC YEAR 2025 AND ONWARDS

Web Resources

https://studylib.net/doc/10455461/simple-harmonic-motion--shm-kx-mv

https://www.youtube.com/watch?v=5H8aRCyEGnU

Suggested Activities and/ or Formative Assessment

Activity 1

Bring pendulum apparatus in the classroom and ask students to measure its time period. Also use this apparatus to show the factors affecting time period of pendulum.

Activity 2

Assign any of the topics to each student or in groups and ask them to present in front of classmates. (It will be more productive if students use presentation making tools e.g., PowerPoint, Canvas etc. for this purpose)".

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resources%20Guide%202025%20HSSC%20Physics.pdf

Further Resource

For additional resources related to teaching, learning and formative assessments, please refer to **Learn Smart Classroom by Knowledge Platform:**

https://akueb.knowledgeplatform.com/login

FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
8. Waves	14

Sub-Topic	Range of SLOs	Periods (40 mins)
8.1 Wave Motion	8.1.1 – 8.1.4	02
8.2 Transverse and Longitudinal Waves	8.2.1, 8.2.2	02
8.3 Superposition of Waves	8.3.1 – 8.3.4	02
8.4 Stationary Waves	8.4.1 – 8.4.3	04
8.5 Doppler's Effect	8.5.1 – 8.5.5	03
8.6 Waves in modern world	8.6.1	01

FOR ACADEMIC YEAR 2025 AND ONWARDS

Web Resources

https://www.youtube.com/watch?v=UDyhcxyR_90

https://www.youtube.com/watch?v=yd-G6KYwzvA

https://www.youtube.com/watch?v=-_xZZt99MzY

http://www.sengpielaudio.com/calculator-airpressure.htm

https://www.youtube.com/watch?v=CAe3IkYNKt8

https://www.youtube.com/watch?v=_S7-PDF6Vzc

Standing Waves Demonstration
https://www.youtube.com/watch?v=5-0U4-Sd4EE

Standing waves in closed pipe
https://www.youtube.com/watch?v=h4OnBYrbCjY

Suggested Activities and/ or Formative Assessment

Activity 1

Divide some topics among students in groups and ask them to discuss and make notes. Now give them some tasks on the relevant topics to conclude their understanding.

Activity 2

Few sample questions for assessments.

https://www.slideshare.net/RamC3/physics-notes-solved-numerical-of-physics-first-year

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resources%20Guide%202025%20HSSC%20Physics.pdf

Further Resource

For additional resources related to teaching, learning and formative assessments, please refer to **Learn Smart Classroom by Knowledge Platform:**

https://akueb.knowledgeplatform.com/login

FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
9. Physical Optics	17

Sub-Topic	Range of SLOs	Periods (40 mins)
9.1 Nature of Light	9.1.1 – 9.1.3	02
9.2 Interference of Light	9.2.1 – 9.2.6	04
9.3 Michelson's Interferometer	9.3.1 – 9.3.4	03
9.4 Diffraction of Light	9.4.1 – 9.4.4	02
9.5 Bragg's Law	9.5.1 – 9.5.4	02
9.6 Polarisation	9.6.1 – 9.6.7	04

FOR ACADEMIC YEAR 2025 AND ONWARDS

Web Resources

https://www.youtube.com/watch?v=cwlmCx83K5I

https://www.youtube.com/watch?v=-mNQW5OShMA

Suggested Activities and/ or Formative Assessment

Activity 1

Few sample questions for assessments.

https://www.slideshare.net/RamC3/physics-notes-solved-numerical-of-physics-first-year

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resources%20Guide%202025%20HSSC%20Physics.pdf

Further Resource

For additional resources related to teaching, learning and formative assessments, please refer to **Learn Smart Classroom by Knowledge Platform:**

https://akueb.knowledgeplatform.com/login

FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
10. Thermodynamics	25

Sub-Topic	Range of SLOs	Periods (40 mins)
10.1 Heat and Temperature	10.1.1 – 10.1.4	02
10.2 Fundamentals of the Ideal Gas	10.2.1, 10.2.2	02
10.3 Kinetic Molecular Theory of Gases	10.3.1 – 10.3.5	02
10.4 Internal Energy and Work	10.4.1 – 10.4.4	02
10.5 First Law of Thermodynamics	10.5.1 – 10.5.3	03
10.6 Molar Specific Heat of Gases	10.6.1 – 10.6.3	02
10.7 Reversible and Irreversible Process	10.7.1	01
10.8 Second Law Of Thermodynamics	10.8.1	02
10.9 Carnot Engine	10.9.1 – 10.9.5	03
10.10 Refrigerator	10.10.1, 10.10.2	03

FOR ACADEMIC YEAR 2025 AND ONWARDS

10.11 Entropy	10.11.1 – 10.11.5	03

Web Resources

https://chem.libretexts.org/Under_Construction/Purgatory/Kinetic_Theory_of_Gases

https://www.ck12.org/section/second-law-of-thermodynamics-::of::-thermodynamics-::of::-ck-12-physics-intermediate-teachers-edition/

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resources%20Guide%202025%20HSSC%20Physics.pdf

Further Resource

For additional resources related to teaching, learning and formative assessments, please refer to **Learn Smart Classroom by Knowledge Platform:**

https://akueb.knowledgeplatform.com/login

FOR ACADEMIC YEAR 2025 AND ONWARDS

Note: This teacher-led pacing guide has been developed by educators from AKU-EB affiliated schools to support fellow teachers teaching the AKU-EB syllabi by

- ensuring smooth transition of a school's academic year.
- ensuring curricular continuity in schools.
- predicting the time and pace of syllabi implementation.

This document also contains **suggested activities and/ or formative assessments** that may enhance the learning experience. Please note that these activities are meant to serve as suggestions. As educators, you have the flexibility and autonomy to adapt and modify them to best suit the needs of your students and the dynamics of your classroom.

You are advised to use an ad-blocker while accessing the websites and web resources. In case any website is not functional for any reason, you may inform us at examination.board@aku.edu for an alternative or search material via any search engine.

FOR ACADEMIC YEAR 2025 AND ONWARDS